

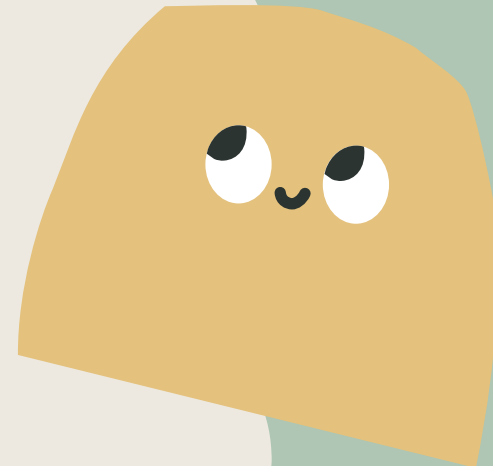
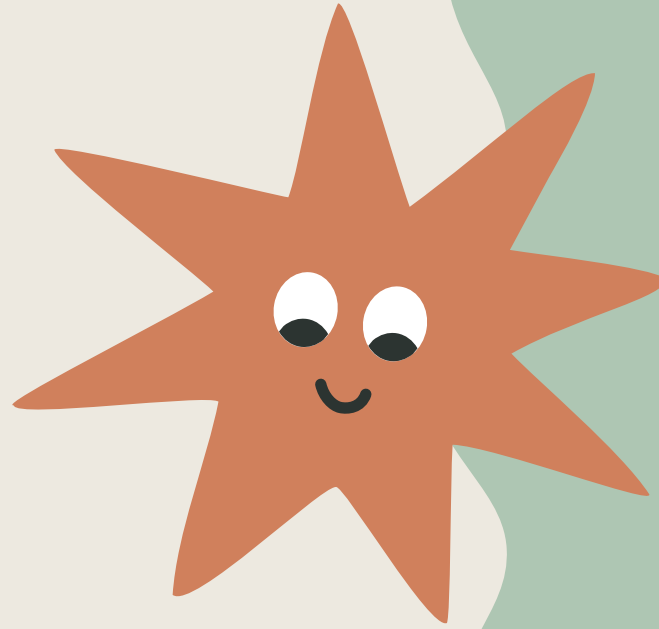
UNIT 2

FUNDAMENTALS OF

OPERATING SYSTEM

9TH COMPUTER SCIENCE
SINDH TEXTBOOK BOARD

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Agenda

Introduction to Operating System

Function of Operating System

Types of Interface

Types of Operating System

Introduction to Operating System

An Operating System is a software which performs all the basic tasks like booting the computer, file management, memory management, process management, and controlling peripheral devices such as hard disk, printer, etc. It manages computer resources efficiently. Most common operating systems are: DOS, Windows, Linux, Android, Mac OS and iOS.

Functions of OS

Booting

Booting is a process of starting the computer operating system. It checks the computer resources and makes it ready to perform different tasks.

Resource Management

Operating system manages all the hardware and software resources. This includes allocation and de-allocation of processor, memory, access to shared memory and access to network resources.

User Interface or Command Interpreter

User Interface or Command Interpreter means to interact with computer via OS in order to get any task done.

Memory Management

Memory management module performs the task of allocation and de-allocation of memory space to programs and data in need of these resources.

Functions of OS

(I/O) Management

An Operating System provides the device driver to facilitate I/O functions involving I/O devices. These device drivers are software that control I/O devices through their controllers.

File Management

It manages all the file-related activities such as organization storage, retrieval, naming, sharing, and protection of files.

Process Management

A process is a job or activity to be performed by the system. Process management manages creation, deletion, suspension and resumption of processes.

User Management

It is the way an operating system controls who can use the computer and what they can do. The OS creates and manages user accounts (like Admin, Guest, or Standard User).

Types of Interface

Command Line Interface (CLI)

A Command Line Interface (CLI) is a text-based representation in which the user types commands at a place called the prompt to operate the computer. CLI is difficult to use because the user has to remember the commands and their syntaxes, but it is fast in use because text mode takes fewer resources. It was primarily provided to users by computer terminals on UNIX, and personal computers including MS-DOS and Apple DOS.

Graphical User Interface (GUI)

A GUI (Graphical User Interface) provides a user-friendly environment where users can interact with computers through graphical objects such as menus, icons, buttons, and other graphical objects. It is easy to use as users are supposed to just click on a picture to run commands without memorizing them. GUI is slower than CLI as graphical mode takes more memory and resources. Windows and iOS are examples of GUI.

Types of Operating System

Single User And Multi-user Operating System

In a Single User Operating System, a single user can access the computer system at a time. DOS for PCs and Windows 98 for PCs are examples of single user operating systems.

A Multi-User Operating System allows multiple users to access the computer at the same time. The operating system manages the memory and resources among the various users according to the requirement. Linux and UNIX are the most common examples of multi-user operating systems.

Batch Processing Operating System

Batch Processing runs programs with minimum human interaction. Users submit jobs to an operator, who groups similar jobs together and runs them as a batch. The system automatically executes jobs one after another. Benefits: shared by multiple users, very less idle time, and the next job starts right after the current one. Examples of batch OS are; IBM OS/360, Unisys MCP & Burroughs MCP.

Modern Batch Tasks & Tools:

Payroll Processing Systems: Software that calculates salaries and prints paychecks for entire companies in a single bulk run at the end of a pay period.

Bank Statement & End-of-Day Reconciliations: Automated routines that process daily transactions and update account balances overnight

Types of Operating System Cont....

Time Sharing Operating System

Time Sharing allows multiple users to share computer resources simultaneously. Each task is given a short time slot called quantum to execute, and the CPU switches rapidly between tasks. This makes all users feel the system is not shared and gives immediate response. These systems are also called Multitasking Systems. Tasks can be from a single user or different users. Examples of Time-Sharing OS are; Linux, Windows Server.

Real Time Processing Operating System (RT OS)

Real Time Processing OS is a time-bound operating system with a fixed time limit. Processing must finish within the defined time, or the system fails. It has two types:

1. Hard Real Time System – Guarantees critical tasks complete on time; no delay acceptable. *Examples:* Missile systems, Air Traffic Control.
2. Soft Real Time System – Deadline is less strict; not compulsory every time. Critical tasks get priority but may finish before a certain time. *Examples:* Online transactions, Airline ticket reservation.